ODE Definitions and Terminology

- An **ODE** relates functions of one variable to their derivatives
  - **Order** is the highest derivative order present
  - **Autonomous** means no explicit dependence on t
  - An **Initial Condition (IC)** specifies $\boldsymbol{y}(t_0) = \boldsymbol{y}_0$
  - An **Initial Value Problem (IVP)** is an ODE with initial conditions
  - A **Solution** satisfies the ODE and IC on an interval containing t_0
- $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{b}(t)$ is a **Linear ODE** if the coefficients are independent of $\boldsymbol{y}$.
  - **Constant Coefficient (CC)** means $\boldsymbol{A}$ and $\boldsymbol{b}$ do not depend on t
  - $\boldsymbol{b}(t)$ is the **Forcing**. If $\boldsymbol{b}(t) = \boldsymbol{0}$ the ODE is **Homogeneous**.

- An **Equilibrium Solution** of $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$ is constant $\boldsymbol{y}(t) = \boldsymbol{y}_*$ with $\boldsymbol{f}(\boldsymbol{y}_*) = \boldsymbol{0}$

Separation of Variables

- A scalar ODE is **Separable** if it can be written

$$y' = \frac{dy}{dt} = g(t)h(y)$$

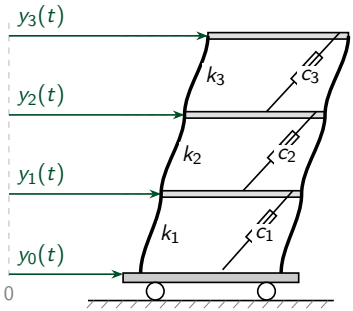
  - Multiply by dt, divide by $h(y)$, **Separate Variables** and integrate

$$dy = g(t)h(y)\,dt \quad \Longrightarrow \quad \frac{dy}{h(y)} = g(t)\,dt \quad \Longrightarrow \quad \int \frac{dy}{h(y)} = \int g(t)\,dt + C$$

  - If there is an initial condition, use it to determine C.

Three-Story Building

Displacements: $\boldsymbol{y}(t) = (y_1(t), y_2(t), y_3(t))$ (m), t in seconds.



$m_3 = 2\,\mathrm{kg}$
$k_3 = 2\,\mathrm{N/m}$
$c_3 = 1\,\mathrm{N\,s/m}$

$m_2 = 3\,\mathrm{kg}$
$k_2 = 3\,\mathrm{N/m}$
$c_2 = 1\,\mathrm{N\,s/m}$

$m_1 = 4\,\mathrm{kg}$
$k_1 = 4\,\mathrm{N/m}$
$c_1 = 2\,\mathrm{N\,s/m}$

$y_0(t) = \sin(t)\ \mathrm{m}$

Elastic and Damping Forces Between Floors

1. Displacements $y_i(t)$ in m rightwards from origin.
2. Between floors $i-1$ and i:

$$\underbrace{y_i - y_{i-1}}_{\text{relative displacement } [\mathrm{m}]} \qquad \underbrace{y_i' - y_{i-1}'}_{\text{relative velocity } [\mathrm{m/s}]}$$

3. Walls transmit forces

$$f_i = \underbrace{k_i(y_i - y_{i-1})}_{\text{elastic}} + \underbrace{c_i(y_i' - y_{i-1}')}_{\text{damping}} \quad [\mathrm{N}]$$

4. Walls exert forces $-f_i$ on floor i and $+f_i$ on floor $i-1$.
5. Level 0 is the driven base. Its walls exert force on floor 1

$$-f_1 = 4(y_0 - y_1) + 2(y_0' - y_1') \quad [\mathrm{N}].$$

Force Units Newton says $f = ma$

1. Spring force in N is stiffness (k_i $\mathrm{N/m}$) times relative displacement ($y_i - y_{i-1}$ m)

$$k_i(y_i - y_{i-1}): \qquad \frac{\mathrm{N}}{\cancel{\mathrm{m}}}\cancel{\mathrm{m}} = \mathrm{N}$$

2. Damping force in N is damping (c_i $\mathrm{N\,s/m}$) times relative velocity ($y_i' - y_{i-1}'$ $\mathrm{m/s}$)

$$c_i(y_i' - y_{i-1}'): \qquad \frac{\mathrm{N}\,\cancel{\mathrm{s}}}{\cancel{\mathrm{m}}}\frac{\cancel{\mathrm{m}}}{\cancel{\mathrm{s}}} = \mathrm{N}$$

3. Force in N is mass (m_i kg) times acceleration (y_i'' $\mathrm{m/s^2}$)

$$m_i y_i'': \qquad \mathrm{kg}\,\frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

Floor Balances

1. Net force in N is mass times acceleration

$$4y_1'' = -f_1 + f_2, \qquad 3y_2'' = -f_2 + f_3, \qquad 2y_3'' = -f_3$$

2. Use the displayed stiffness and damping values

$$\begin{aligned} f_1 &= 4(y_1 - y_0) + 2(y_1' - y_0') &&= 4y_1 - 4y_0 + 2y_1' - 2y_0' \\ f_2 &= 3(y_2 - y_1) + (y_2' - y_1') &&= 3y_2 - 3y_1 + y_2' - y_1' \\ f_3 &= 2(y_3 - y_2) + (y_3' - y_2') &&= 2y_3 - 2y_2 + y_3' - y_2' \end{aligned}$$

3. Collect floor terms on the left and base terms on the right

$$\begin{aligned} 4y_1'' + 3y_1' - y_2' + 7y_1 - 3y_2 &= 2y_0' + 4y_0 \\ 3y_2'' - y_1' + 2y_2' - y_3' - 3y_1 + 5y_2 - 2y_3 &= 0 \\ 2y_3'' - y_2' + y_3' - 2y_2 + 2y_3 &= 0 \end{aligned}$$

Floor Equations Continue

$$\begin{aligned} 4y_1'' + 3y_1' - y_2' + 7y_1 - 3y_2 &= 2y_0' + 4y_0 \\ 3y_2'' - y_1' + 2y_2' - y_3' - 3y_1 + 5y_2 - 2y_3 &= 0 \\ 2y_3'' - y_2' + y_3' - 2y_2 + 2y_3 &= 0 \end{aligned}$$

$$\implies \begin{array}{llll} 4y_1'' & + 3y_1' - y_2' & + 7y_1 - 3y_2 & = 2y_0' + 4y_0 \\ 3y_2'' & - y_1' + 2y_2' - y_3' & - 3y_1 + 5y_2 - 2y_3 = & 0 \\ 2y_3'' & - y_2' + y_3' & - 2y_2 + 2y_3 = & 0 \end{array}$$

$$\implies \begin{array}{llll} 4y_1'' + 0y_2'' + 0y_3'' & + 3y_1' - y_2' + 0y_3' & + 7y_1 - 3y_2 + 0y_3 = 2y_0' + 4y_0 \\ 0y_1'' + 3y_2'' + 0y_3'' & - y_1' + 2y_2' - y_3' & - 3y_1 + 5y_2 - 2y_3 = & 0 \\ 0y_1'' + 0y_2'' + 2y_3'' & + 0y_1' - y_2' + y_3' & + 0y_1 - 2y_2 + 2y_3 = & 0 \end{array}$$

Second-Order Matrix System

- Force balance is

$$\boxed{\boldsymbol{M}\boldsymbol{y}'' + \boldsymbol{C}\boldsymbol{y}' + \boldsymbol{K}\boldsymbol{y} = \boldsymbol{b}(t)}$$

where

- $\boldsymbol{M}$, $\boldsymbol{C}$, and $\boldsymbol{K}$ have units kg, $\mathrm{N\,s/m}$, and $\mathrm{N/m}$ respectively

$$\underbrace{\boldsymbol{M} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}}_{\text{mass matrix}} \quad \underbrace{\boldsymbol{C} = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}}_{\text{damping matrix}} \quad \underbrace{\boldsymbol{K} = \begin{pmatrix} 7 & -3 & 0 \\ -3 & 5 & -2 \\ 0 & -2 & 2 \end{pmatrix}}_{\text{stiffness matrix}}$$

- The external force vector $\boldsymbol{b}(t)$ has units N

$$\boldsymbol{b}(t) = \begin{pmatrix} 2y_0'(t) + 4y_0(t) \\ 0 \\ 0 \end{pmatrix}$$

First-Order System (Six Scalar Equations)

1. Define velocity $\boldsymbol{v} = \boldsymbol{y}'$ in $\mathrm{m/s}$. Stack displacement and velocity into the six vector

$$\boldsymbol{z} = \begin{pmatrix} \boldsymbol{y} \\ \boldsymbol{v} \end{pmatrix} \qquad \boldsymbol{z}' = \begin{pmatrix} \boldsymbol{y}' \\ \boldsymbol{v}' \end{pmatrix}$$

2. Isolate acceleration, then multiply by $\boldsymbol{M}^{-1}$

$$\boldsymbol{M}\boldsymbol{y}'' = -\boldsymbol{K}\boldsymbol{y} - \boldsymbol{C}\boldsymbol{y}' + \boldsymbol{b}(t)$$

$$\boldsymbol{y}' = \boldsymbol{v} \qquad \boldsymbol{v}' = \boldsymbol{y}'' = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y} - \boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v} + \boldsymbol{M}^{-1}\boldsymbol{b}(t)$$

3. The top three equations give velocity. The bottom three give acceleration

$$\boldsymbol{z}' = \begin{pmatrix} \boldsymbol{y}' \\ \boldsymbol{v}' \end{pmatrix} = \underbrace{\begin{pmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{M}^{-1}\boldsymbol{K} & -\boldsymbol{M}^{-1}\boldsymbol{C} \end{pmatrix}}_{\boldsymbol{A}} \underbrace{\begin{pmatrix} \boldsymbol{y} \\ \boldsymbol{v} \end{pmatrix}}_{\boldsymbol{z}} + \begin{pmatrix} \boldsymbol{0} \\ \boldsymbol{M}^{-1}\boldsymbol{b}(t) \end{pmatrix}$$

Here $\boldsymbol{0}$ and $\boldsymbol{I}$ are the 3×3 zero and identity matrices.

Acceleration Units

- Acceleration in $\mathrm{m/s^2}$ is force (N) divided by mass (kg)

$$\frac{\mathrm{N}}{\mathrm{kg}} = \frac{\cancel{\mathrm{kg}}\,\mathrm{m}}{\mathrm{s}^2\,\cancel{\mathrm{kg}}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

- Every term in $\boldsymbol{v}' = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y} - \boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v} + \boldsymbol{M}^{-1}\boldsymbol{b}$ has these units

$$\underbrace{\frac{1}{\cancel{\mathrm{kg}}}\frac{\cancel{\mathrm{kg}}}{\mathrm{s}^2}\,\mathrm{m}}_{\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y}} = \frac{\mathrm{m}}{\mathrm{s}^2}, \qquad \underbrace{\frac{1}{\cancel{\mathrm{kg}}}\frac{\cancel{\mathrm{kg}}}{\mathrm{s}}\frac{\mathrm{m}}{\mathrm{s}}}_{\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

First-Order Building IVP

$$\boldsymbol{z}' = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{r}(t) = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -7/4 & 3/4 & 0 & -3/4 & 1/4 & 0 \\ 1 & -5/3 & 2/3 & 1/3 & -2/3 & 1/3 \\ 0 & 1 & -1 & 0 & 1/2 & -1/2 \end{pmatrix} \boldsymbol{z} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ y_0 + y_0'/2 \\ 0 \\ 0 \end{pmatrix}$$

- Initial positions ($\boldsymbol{y}(0)$ m) and velocities ($\boldsymbol{v}(0)$ $\mathrm{m/s}$) give

$$\boldsymbol{z}(0) = \begin{pmatrix} \boldsymbol{y}(0) \\ \boldsymbol{v}(0) \end{pmatrix}$$

- Prescribing base motion $y_0(t)$ and the six values for $\boldsymbol{z}(0)$ completes the IVP.